Week2

Q1 How many bits are required to store a text document consisting of 8000 alphanumeric characters using 8-bit ASCII?

A. 32000 bits
B. 64000 bits
C. 16000 bits
D. 42000 bits

8bit ASCII>>> each character is represented by 8 bits

ASCII - 7 bit old
ASCII - 8 bit new
UCS (8 bits)
UCS-2 (2 bytes i.e 16 bits) 1 byte= 8bits
UCS-4 (4 bytes 32bits)

Size of file = no of characters x size of each character
= 8000 x 8
= 64000

1 bit 0,1 2 distinct a, A a-0, A-1
a,b,c,d 4 diff combo
N bit gives 2^N distinct combinations
4combo 2^2= 4 , 2bits text = aabbccddd

a -00, b 01, c-10, d-11

Q2: A certain text document consisting of only alphanumeric characters (including spaces) takes 1,36,000 bits, when encoded with UCS-4 (32 bit) encoding. How many bits will the same document take if encoded with ASCII 7-bit encoding?

(Concept: Encoding efficiency)

A. 34000 bits
B. 68000 bits
C. 29750 bits
D. 136000 bits

Answer: C

Size of file = no of characters x size of each character
Case 1: UCS 4
No of characters = size of file / size of each character
= 1,36,000 bits/32 bit
= 4250 characters
Case 2: ASCII 7 bit
Size of file = no of characters x size of each character
= 4250 x 7
= 29750 bits

Q3. Let `L = {'a', 'b', 'c', 'd', 'A', 'B', 'C', 'D', '0', '1', ' '}` be a complete character set (i.e., only these characters can be used to represent text in the document). If a document that uses fixed encoding for all characters is created using the character set L and has a disk size of 2 Kilobytes, the number of characters in the documents would be_____. [Take 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 Kilobytes and so on.]

[MCQ: 3 points]

A. 2000
B. 4000
C. 8000
D. 16000

Answer: B
11 diff combo
3 bits 2^3 = 8 distinct combinations>> 8 different characters
4 bits 2^4 = 16 distinct combinations (a= 1000 A= 1100)
No of characters = (size of file/ bits required to store each character)
= (2 x 1000 x 8bits/ 4 bits)
= 4000 characters

2bit : 00,01,10,11 2^2=4 a,b,c,d aaabbbccdd

1character will require 4bit, n characters = nx 4bits = size of file

Q4. Which of the following is a correct way to create a hyperlink in HTML?

A. <a href = “www.onlinedegree.iitm.ac.in”> IITM Online Degree Website </a>
B. <a url = “www.onlinedegree.iitm.ac.in” IITM Online Degree Website /a>
C. <a link = “www.onlinedegree.iitm.ac.in”> IITM Online Degree Website </a>
D. <a> www.onlinedegree.iitm.ac.in <IITM Online Degree Website /a>

Q5. Consider the following code segment.

HTML code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
    <link rel="stylesheet" href="style.css">
    <style>
        div{
            color:blue
        }
    </style>

</head>
<body>
    <div style="color: red;">IITM 1</div>
</body>
</html>
```

CSS code

```
div{color:yellow}
```

What will be the color of text IITM 1?

! Important>>Inline>> Internal>>External

!important has highest precedence over evrything

!important> inline>id>class>element>internal>external

Ans: It will take red color

6) Consider the following code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
    <style>
        #iitm{color: red}
        .iitm{color:green}
        div{color:yellow}
   </style>
</head>
</head>
<body>
   <div class = "iitm" id="iitm" >IITM Online</div>
</body>
</html>
```

What will be the color of text "IITM Online" if the above styling is used?

- ○ Black
- ○ Red
- ○ Green
- ○ Yellow

Ans: red color
Element- department, classes-class, student-id
Based on selector:
! important>>>id>>class>> element

Q7. Create a simple HTML document with the following requirements:

[MCQ: 2 points]

1. Include a heading with the text `"My Web Page"` using an `<h1>` tag.
2. Add a paragraph with the text `"Welcome to the world of HTML!"` using a `<p>` tag.
3. Create an unordered list with three list items. The list items should contain the text `"Item 1"`, `"Item 2"` and `"Item 3"` respectively, using `<li>` tags.
4. Include a link to `"https://www.example.com"` with the anchor text `"Visit Example.com"` using an `<a>` tag.

Which of the following HTML code is correct to fulfill these requirements?

A.

```
<body>
    <h1>My Web Page</h1>
    <p>Welcome to the world of HTML!</p>
    <ol>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ol>
    <a href="https://www.example.com">Visit Example.com</a>
</body>
```

B.

```
<body>
    <h1>My Web Page</h1>
    <p>Welcome to the world of HTML!</p>
    <ol>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ol>
   <a href="Visit Example.com">https://www.example.com</a>
</body>
```

C.

```
<body>
    <h1>My Web Page</h1>
    <p>Welcome to the world of HTML!</p>
    <ul>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ul>
    <a href="Visit Example.com">https://www.example.com</a>
</body>
```

D.

```
<body>
    <h1>My Web Page</h1>
    <p>Welcome to the world of HTML!</p>
    <ul>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ul>
    <a href="https://www.example.com">Visit Example.com</a>
</body>
```

Answer: D

Q8: Which of the following is an incorrect way of creating an ordered list in HTML?

A.

```
<ol type = "1">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

B.

```
<ol type = "a">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

C.

```
<ul type = "i">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ul>
```

D.

```
<ol>
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

Answer: C

Q9: Consider the below HTML and CSS styling.

```
<html>
    <head>
        <style>
            h3 {
                color: green;
            }
            .content {
                color: red;
            }
        </style>
    </head>
    <body>
        <h3 class = "content">This is the main Content</h3>
        <button onclick = "remove_class"> Click Me </button>
    </body>
</html>
```

Assuming clicking the button with the text “Click Me” removes the class “content” from the “h3” element, what will be the text colour of the text placed inside the “h3” element, before and after clicking the button with the text “Click Me” (Assume a normal browser is used which shows black text on white background)?

A. Before: Red
   After: Green

B. Before: Red
   After: Black

C. Before: Green
   After: Black

D. Before: Green
   After: Red

Answer: A

Q10: Consider the following HTML document.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #mybox{
            border-style: dashed;
            background-color: orange;
        }
        .boxes{
            border-style: solid;
            border-color: blue;
        }
        div{
            width: 200px;
            border: 4px dotted yellow;
        }
    </style>
</head>
<body>
    <div id="mybox" class="boxes" style="background-color: limegreen;">
        <h1>Hello MAD-I</h1>
    </div>
</body>
</html>
```

How will the browser render the above HTML document?

A.

B.

C.

D.

Answer: C

Q11. An HTML code and CSS code is given below. Which of the following correctly represents its rendered output?

CSS Code:

```
#one{color: blue;}
.two{color: red !important;}
#two{color: green}
#three{color: green;}
```

HTML Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Document</title>
    <link href="style.css" rel="stylesheet">
    <style>
        body{font-weight: bold;}
        p{color: violet !important ;}
    </style>
</head>
<body>
    <span id="one">Content 1</span>
    <span class="two" id="two" >Content 2</span>
    <p id="three">Content 3</p>
</body>
</html>
```

A.

Content 1 Content 2

Content 3

B.

Content 1

Content 2

Content 3

C.

Content 1 Content 2

Content 3

D.

Content 1 Content 2

Content 3

Answer: C